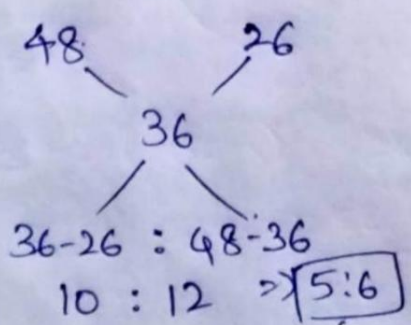
Ex:3:-[B.Q:-3]:- What Quantity of Sugar of Superior quality costing 48/- per kg should a shopkeeper mix with 30 kg sugar of low quality costing 26/- per kg, so that he makes a profit of 25% on selling the mixture @ 45/- per kg?

Ans:- When Dearer & Cheaper Cost prices given we have to Consider Mean price as also Cost price only. But in this question Mean S.P given, we have to find Mean C.P from it.

$$1.25\,CP_m = 45 \Rightarrow CP_m = \frac{45}{1.25} = \frac{\cancel{45}^{9}}{\cancel{5}} \times \frac{4}{\cancel{5}} = 36/-$$



Low costing 30kgs
⇒ 6 parts = 30 kgs
⇒ 1P = 5kgs
Superior Quantity = 5 parts
= 5×5 = 25kgs

Ex:4:-[B.Q:-4]:- A Container has 30 ltrs of water. If 3 litres of water is replaced by 3 litres of spirit & this operation is repeated twice, what will be the quantity of water in new mixture?

Ans:- According to our Conclusion on Ratios.

$$W_f = W_i\left(1 - \text{Replacement}\right)^n$$

$$= 30\left(1 - \frac{3}{30}\right)^3 = 30\left(1 - \frac{1}{10}\right)^3$$

$$= \cancel{30}^{3} \times \frac{9}{\cancel{10}} \times \frac{9}{10} \times \frac{9}{10} = 27 \times 0.9 \times 0.9$$

$$= 27 \times 0.81 = 21.87 \text{ ltrs.}$$

Ex:5:- [B.Q:-7]:- A, B, C & D enter into partnership investing A $\frac{1}{3}$rd of, B $\frac{1}{4}$th, C $\frac{1}{5}$th & D rest of the capital. How much share did A gets on a profit of 2460/-?

Ans:- Ratio of P/L = Ratio of Investments in simple partnership

$$= \frac{1}{3} : \frac{1}{4} : \frac{1}{5} : \left(1 - \left(\frac{20+15+12}{60}\right)\right)$$

$$= \frac{1}{3} : \frac{1}{4} : \frac{1}{5} : \frac{13}{60} = \left[\frac{1}{3} : \frac{1}{4} : \frac{1}{5} : \frac{13}{60}\right] 60$$

$$= 20 : 15 : 12 : 13$$

$$A \text{ profit} = \frac{20}{60} \times \frac{2460}{41} = 820/-.$$

Ex:6:- [B.Q:-8]:- A & B entered into a partnership investing 25k & 30k respectively. After 4 months 'C' also joined the business with 35k investment. What's share of C on annual profit of 47k?

Ans:- Its a Compound partnership [T.P of investment is diff]

P/L Ratio = $I_1T_1 : I_2T_2 : I_3T_3 : \cdots$

Here total time of business is 1 year = 12 months

P/L Ratio = $25k \times 12 : 30k \times 12 : 35k \times (12-4)$

$= 5 \times 3 : 6 \times 3 : 7 \times 2 = 15 : 18 : 14$

C's share = $\frac{14}{47} \times 47k = \boxed{14k}$

Ex:7:- [B.Q:-9]:- A & B enter into a partnership with capitals as 7:9. At the end of 8 months, A withdraws. If they receive the profits in the ratio 8:9. Find how long B's capital was used?

Ans:- Its Complex Partnership

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P/L Ratio = $I_1 t_1 : I_2 t_2$ ... ②

8 : 9 = 7x8 : 9x t_2

①

from ① ⇒ 8 parts = 8x7 ⇒ 1 part = 7.

from ② ⇒ 9t = 9 parts = 9x7 ⇒ t_2 = 7 months

Ex: 8:- [BIQ:-11]:- A, B & C started the Business with their investments in the ratio 1:3:5. After 4 months, A invested the same amount as before & B as well as C withdrew half of their investments. The ratio of their profits @ the end of year is?

Ans:- It is compound Partnership & time period of Business is 1 year (or) 12 months.

P/L Ratio = $I_1 t_1 : I_2 t_2 : I_3 t_3$: - - -

Investments ratio = 1:3:5 ⇒ x, 3x, 5x.

P/L Ratio = $4x x + 8x(x+x) : 4x3x + 8x\left(3x - \frac{3x}{2}\right) : 4x5x + 8x\left(5x - \frac{5x}{2}\right)$

= 20x : 24x : 40x = **5 : 6 : 10**